PRODUCT USER MANUAL

For Mediterranean Sea L4 and L3S SST Reprocessed Products

SST_MED_SST_L4_REP_OBSERVATIONS_010_021
SST_MED_PHY_L3S_MY_010_042
Issue: 1.0

Contributors : Andrea Pisano, Claudia Fanelli, Andrea Massi, Cristina Tronconi, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno Nardelli, Daniele Ciani

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

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a) Structure and semantic of NetCDF maps files..... **Erreur ! Signet non défini.**

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
MED	Mediterranean Sea
GHRSST	Group for High Resolution SST
REP	Reprocessed
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document provides practical information on the Multi-Year (MY), also known as Reprocessed (REP), operational production of satellite-based Sea Surface Temperature data over the Mediterranean Sea (MED) reprocessed (REP) L4 SST products (Table 1). These data are produced by the Italian National Research Council – Institute of Marine Sciences (CNR-ISMAR) and are particularly suitable for climate applications.

Table 1. *Copernicus Marine Service - Mediterranean Sea Reprocessed L4 and L3S SST data.*

PRODUCT REFERENCE	FULL NAME	AREA	TYPE	LEVEL
SST_MED_SST_L4_REP_ OBSERVATIONS_010_021	Mediterranean Sea - High Resolution L4 Sea Surface Temperature Reprocessed	MED	REP	L4
SST_MED_PHY_L3S_MY_010_042	Mediterranean Sea - High Resolution L3S Sea Surface Temperature Reprocessed	MED	REP	L3S

These products are a collection of daily (nighttime), merged multi-sensor (Level-3S, L3S) and optimally interpolated (Level-4, L4) satellite-based estimates of the foundation (namely, the temperature free, or nearly-free, of any diurnal cycle), SST over the Mediterranean Sea (MED), at 0.05° grid covering the period from January 1st 1982 to present (currently, up to one month before real time).

The MED SST products are built from a consistent reprocessing of the collated Level-3C (merged single-sensor, L3C) climate data record (v.3.0) provided by the ESA Climate Change Initiative (CCI) (Embury et al., 2024) and the Copernicus Climate Change Service (C3S) initiatives, but also include in input an adjusted version of the AVHRR Pathfinder dataset version 5.3 (Saha et al., 2018) to increase the input observation coverage.

These products replace the previous reprocessed versions, which were based on an earlier version (v.2.1) of the ESACCI/C3S SST data.

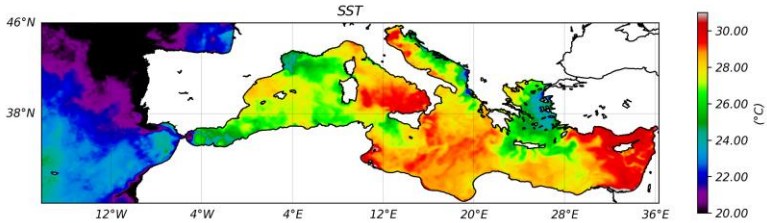
2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The Copernicus Marine Mediterranean Sea Reprocessed SST datasets (MED-REP-L3S, MED-REP-L4) consist of daily (nighttime), merged multi-sensor (L3S) and optimally interpolated (L4) satellite-based estimates of the foundation SST in the Mediterranean Sea and adjacent North Atlantic box over a 0.05° resolution grid, covering the period from January 1st 1982 to present. This product is built from a consistent reprocessing of the collated level-3 (merged single-sensor, L3C) climate data record (v.3.0) provided by the ESA Climate Change Initiative (CCI) and the Copernicus Climate Change Service (C3S) initiatives, but also include in input an adjusted version of the AVHRR Pathfinder dataset version 5.3 to increase the input observation coverage.

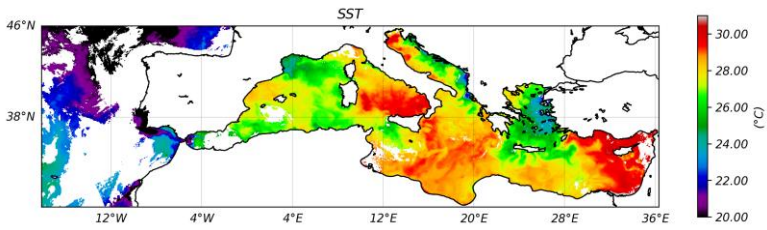
This time series is regularly extended in time (possibly up to one month before real time) based on the availability of the upstream L3C data. The quality and accuracy of this product have been evaluated by comparison with in situ data and can be found in the corresponding QUID (Pisano et al., 2025).

Table 2. Copernicus Marine Service - Mediterranean Sea Reprocessed L4 SST data

Product Lines	SST_MED_SST_L4_REP_OBSERVATIONS_010_021;
Geographical coverage	Mediterranean Sea + adjacent North Atlantic Box:  18.125°W-36.325°E; 30.125°N – 46.025°N
Variables	• Analysed Sea Surface Temperature • Analysis error
Product Type	Reprocessed – L4
Available time series	01/01/1982 - 1 month before real time
Temporal resolution	Daily (Nighttime)
Update Frequency	Daily
Target delivery time	12:00 UTC
Horizontal resolution	0.05° x 0.05° ; regular grid ; lon x lat=1089 x 318
Number of vertical levels	Surface Product

Format	NetCDF4 - CF-1.4 convention compliant

Table 3. Copernicus Marine Service - Mediterranean Sea Reprocessed L3S SST data.

Product Lines	SST_MED_PHY_L3S_MY_010_042
Geographical coverage	Mediterranean Sea + adjacent North Atlantic Box:  18.125°W-36.325°E; 30.125°N – 46.025°N
Variables	• Adjusted Sea Surface Temperature
Product Type	Reprocessed – L3S
Available time series	01/01/1982 - 1 month before real time
Temporal resolution	Daily (Nighttime)
Update Frequency	Daily
Target delivery time	12:00 UTC
Horizontal resolution	0.05° x 0.05°, regular grid ; lon x lat=1089 x 318
Number of vertical levels	Surface Product
Format	NetCDF4 - CF-1.4 convention compliant

b) Production System Description

The regional reprocessed products provide daily (nighttime), merged multi-sensor (L3S) and optimally interpolated (L4) satellite-based estimates of the foundation SST over the Mediterranean Sea (MED), respectively, at 0.05° grid covering the period from January 1st 1982 to present. "Present" refers to approximately one month behind real time, though this timeliness depends on the effective availability of upstream data (see Figure 1).

The MED SST products are built from a consistent reprocessing of the collated Level-3 (merged single-sensor, L3C) climate data record (v.3.0) provided by the ESA Climate Change Initiative (CCI) (Embury et al., 2024) and the Copernicus Climate Change Service (C3S) initiative, but also include in input an adjusted version of the AVHRR

Pathfinder dataset version 5.3 (Saha et al., 2018) to increase the input observation coverage.

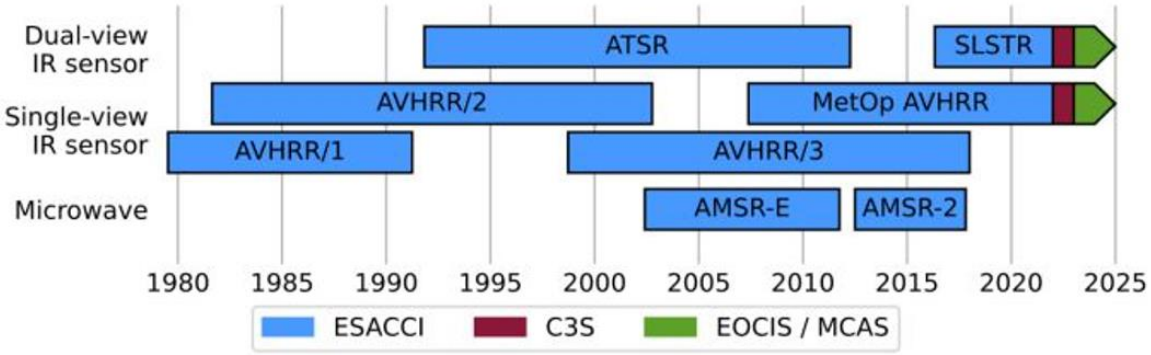


Figure 1. *Overview of products included in CCI SST CDR v3.0. Data from 2022 onwards are Interim-CDR (ICDR) funded by C3S (2022) and EOCIS / MCAS (2023 onwards).*

The interpolation of the L3C upstream data to L4 has been carried out by adapting CNR-ISMAR near-real-time (NRT) SST processing chain (Buongiorno Nardelli et al., 2013) following the approach described in Pisano et al. (2016). The operational chain is organized in various modules/packages, which are managed through a specific System Controller, as specified in the QUID (Pisano et al., 2025).

The upstream data for the REP processing includes collated (merged) daily images (L3C), and have been developed as part of the ESA CCI project (Embury et al. 2024). The ESA CCI Climate Data Record (CDR) covers the period from 1980 to 2021, and has been extended as an Interim CDR (ICDR) since 2022, providing ongoing updates with a short delay of approximately one month. Both SST CDRs and ICDRs are produced using software and algorithms developed through the ESA SST CCI project. CDR v.3 is available at the Centre for Environmental Data Analysis (CEDA[1]), while ICDR v.3 at the UK Group Workspace[2]. These data are produced using different satellite infrared and microwave sensors (Figure 1), i.e. the AVHRR (Advanced Very High Resolution Radiometer), (A)ATSR (Advanced Along-Track Scanning Radiometer), SLSTR (Sea and Land Surface Temperature Radiometer) and AMSR (Advanced Microwave Scanning Radiometer) series (Embury et al., 2024). The resulting data (Table 4) provides a stable, low-bias climate data record.

Following the public release of CCI SST v.3.0 in April 2024, a full reprocessing of the MED data has been carried out. For this purpose, L3C maps derived from infrared sensors only, providing daily (nighttime) SST at 20 cm depth on a 0.05° x 0.05° latitude-longitude grid, have been used as input data. No microwave data were used. To note that ATSR data have not changed since the previous version, so v2.1 data have been used for the whole period available.

[1] https://data.ceda.ac.uk/neodc/eocis/data/global_and_regional/sea_surface_temperature/CDR_v3

[2] https://gws-access.jasmin.ac.uk/public/cds_c3s_sst/data/ICDR_v3/

Table 4. Input upstream L3C data used in the interpolation of the Copernicus Marine Service Mediterranean Sea SST products.

Type	Version	Date range	Region	Data levels
CDR	ESA CCI v.3.0	1982 – 2021	Global	L2P, L3U, L3C
ICDR	ESA CCI v.3.0	2022 - 1 month behind real time	Global	L3C
Pathfinder	v.5.3	1982-2014	Global	L3C

c) Dataset details

Table 5. *List of the four datasets that make up the SST_MED_SST_L4_REP_OBSERVATIONS_010_021 product*

SST_MED_SST_L4_REP_OBSERVATIONS_010_021
Dataset: cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021. This dataset provides daily (nighttime) optimally interpolated (L4) maps of foundation SST over the Mediterranean Sea at 0.05° x 0.05° deg. resolution
analysed_sst [K] analysed_sst:long_name = "analysed sea surface temperature" ; analysed_sst:standard_name = "sea_surface_temperature" ; Mediterranean Reprocessed L4 SST analysis
analysis_error [K] analysis_error:long_name = "estimated error standard deviation of analysed_sst" ; Error standard deviation of L4 SST analysis.
lat [degrees north] lat:long_name = "latitude"; lat:standard_name = "latitude";
lon [degrees east] lon:long_name = "longitude"; lon:standard_name = "longitude";
mask mask:long_name = "sea/land/lake/ice field composite mask"; Land, sea-ice, lake bit mask
Sea_ice_fraction sea_ice_fraction:standard_name = "sea_ice_area_fraction"; sea_ice_fraction:long_name = "sea ice area fraction";

Time[seconds since 1981-01-01 00:00:00] time:long_name = "reference time of sst field"; time:standard_name = "time";

Table 6. *List of the four datasets that make up the SST_MED_PHY_L3S_MY_010_042 product.*

SST_MED_PHY_L3S_MY_010_042
Dataset: cmems_obs-sst_med_phy_my_l3s_P1D-m. This dataset provides daily (nighttime) merged multi-sensor (L3S) maps of foundation SST over the Mediterranean Sea at 0.05°x0.05° deg. Resolution
adjusted_sea_surface_temperature [K] adjusted_sea_surface_temperature :long_name = "adjusted sea surface foundation_temperature"; adjusted_sea_surface_temmperature:standard_name = "sea_surface_foundation_temperature"; Mediterranean Reprocessed L3S adjusted SST
lat [degrees north] lat:long_name = "latitude"; lat:standard_name = "latitude";
lon [degrees east] lon:long_name = "longitude"; lon:standard_name = "longitude";
Time[seconds since 1981-01-01 00:00:00] time:long_name = "reference time of sst field"; time:standard_name = "time";

d) Additional information on some parameters

SST_MED_SST_L4_REP_OBSERVATIONS_010_021

- Real_Value = (Display_Value X scale_factor) + add_offset
- The missing value for this product is: -32768s (analysed_sst and analysis_error)
- Land locations are set to the missing value, which is stored as the "_FillValue" variable attribute in the files.
- Inland water bodies, e.g. small lakes, are masked out.

SST_MED_PHY_L3S_MY_010_042

- Real_Value = (Display_Value X scale_factor) + add_offset
- The missing value for this product is: -32768s (adjusted_sea_surface_temperature)

- Land locations are set to the missing value, which is stored as the "_FillValue" variable attribute in the files.
- Inland water bodies, e.g. small lakes, are masked out.

e) Processing information

The MED REP L4/L3S SST products cover the period from January 1st 1982 to present. Each product is extended in time on a daily basis and currently they reach one month before real time. However, it is worth noting that temporal extension strictly depends on the availability of the upstream L3C data from C3S.

f) Catalogue updating

The MED SST REP products (Table 1) are updated daily by 12:00 (noon) UTC both for L3S and L4 datasets.

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily	Each daily SST field is representative of nighttime SST values (centered at 00:00 UTC)	N/A

h) Product use and impact

For applications such as SST monitoring over climatological time-scales, there are several compelling reasons to prefer the products specifically developed for the Mediterranean Sea over global SST products. The main advantages include

- Use of state-of-the-art algorithms tailored to the Mediterranean region, resulting in higher accuracy compared to global products over the same area.
- This product has sufficient length, accuracy, consistency, and stability to support climate studies and applications, such as estimating SST trends, defining climatologies, and computing anomalies.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [KB]
cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021	<date>000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_REP-MED-v02.0-fv03.0.nc	330
cmems_obs-sst_med_phy_my_l3s_P1D-m	<date>00000-GOS-L3S_GHRSST-SSTfnd-OISST_HR_REP-MED-v02.0-fv04.0.nc	150

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

5) REFERENCES

Buongiorno Nardelli, B., Tronconi, C., Pisano, A., & Santoleri, R. (2013). High and Ultra-High resolution processing of satellite Sea Surface Temperature data over Southern European Seas in the framework of MyOcean project. Remote Sensing of Environment, 129, 1-16.

Embury, O., Merchant, C. J., Good, S. A., Rayner, N. A., Høyer, J. L., Atkinson, C., ... & Donlon, C. (2024). Satellite-based time-series of sea-surface temperature since 1980 for climate applications. Scientific Data, 11(1), 326.

Pisano, A., Fanelli, C., Tronconi, C., Massi, A., Cesarini, C., La Padula, F., Buongiorno Nardelli, B., Ciani. D. (2025). Quality Information Document, Ref: CMEMS-SST-QUID-010-021-022-041-042

Pisano, A., Buongiorno Nardelli, B., Tronconi, C., & Santoleri, R. (2016). The new Mediterranean optimally interpolated pathfinder AVHRR SST Dataset (1982–2012). Remote Sensing of Environment, 176, 107-116.

Saha, K., Zhao, X., Zhang, H., Casey, K. S., Zhang, D., Baker-Yeboah, S., ... & Relph, J. M. (2018). AVHRR Pathfinder version 5.3 level 3 collated (L3C) global 4km sea surface temperature for 1981–Present. NOAA National Centers for Environmental Information.

6) ANNEX

For dataset cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021

netcdf \20250421000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_REP-MED-v02.0-fv03.0

```
{
dimensions:
        time = 1 ;
        lat = 318 ;
        lon = 1089 ;
variables:
        int time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:calendar = "Gregorian" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:comment = "Nominal time of Level 4 analysis" ;
        float lon(lon) ;
                lon:long_name = "longitude" ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
                lon:valid_min = -20.f ;
                lon:valid_max = 40.f ;
        float lat(lat) ;
                lat:long_name = "latitude" ;
                lat:standard_name = "latitude" ;
                lat:axis = "Y" ;
                lat:units = "degrees_north" ;
                lat:comment = "geographical coordinates, WGS84 projection" ;
                lat:valid_min = 29.f ;
                lat:valid_max = 47.f ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:long_name = "analysed sea surface temperature" ;
                analysed_sst:standard_name = "sea_surface_temperature" ;
                analysed_sst:type = "foundation" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:add_offset = 273.15f ;
                analysed_sst:scale_factor = 0.01f ;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
                analysed_sst:source = "[1982-present] ESA CCI SST v.3.0 L3C product (SST at
0.2m); [1982-2014] Pathfinder-PFV5.3 L3C product (SST skin)" ;
                analysed_sst:comment = "[1982-present] Optimal interpolation (OI) SST
measurements from ESA CCI SST v3.0 and PFV53" ;
```

```
            analysed_sst:_FillValue = -32768s ;
        short analysis_error(time, lat, lon) ;
            analysis_error:long_name = "estimated error standard deviation of
analysed_sst" ;
            analysis_error:units = "kelvin" ;
            analysis_error:add_offset = 0.f ;
            analysis_error:scale_factor = 0.01f ;
            analysis_error:valid_min = 0s ;
            analysis_error:valid_max = 10000s ;
            analysis_error:source = "[1982-present] ESA CCI SST v.3.0 L3C product (SST at
0.2m); [1982-2014] Pathfinder-PFV5.3 L3C product (SST skin)" ;
            analysis_error:comment = "Error obtained by Optimal interpolation (OI)" ;
            analysis_error:_FillValue = -32768s ;
        byte sea_ice_fraction(time, lat, lon) ;
            sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
            sea_ice_fraction:long_name = "sea ice area fraction" ;
            sea_ice_fraction:units = "1" ;
            sea_ice_fraction:add_offset = 0.f ;
            sea_ice_fraction:scale_factor = 0.01f ;
            sea_ice_fraction:valid_min = 0b ;
            sea_ice_fraction:valid_max = 100b ;

            sea_ice_fraction:source_ice = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:comment_ice = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:comment = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:source = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:_FillValue = -128b ;
        byte mask(time, lat, lon) ;
            mask:long_name = "sea/land/lake/ice field composite mask" ;
            mask:units = "1" ;
            mask:flag_values = 1b, 2b, 3b, 4b ;
            mask:flag_meanings = "sea land lake ice" ;
            mask:valid_min = 0b ;
            mask:valid_max = -1b ;
            mask:source = "--------" ;
            mask:comment = "b0:1 = grid cell is open sea water; b1:1 = land is present in
this grid cell; b2:1 = lake surface is present in this grid cell; b3:1 = sea ice is present in
this grid cell; b4-7: reserved for future grid mask data" ;
            mask:_FillValue = -128b ;

// global attributes:
            :Conventions = "CF-1.4 " ;
            :title = "Mediterranean Sea SST Analysis L4, Reprocessed using ESA CCI SST
v.3.0 and PFV53 data, 0.05 deg. daily
(cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021)" ;
            :summary = "[1982-present] Daily gap-free maps (L4) at 0.05deg. x 0.05deg.
horizontal resolution over the Mediterranean Sea. The maps are obtained using
nighttime data extracted from ESA CCI SST v.3.0 and PFV53 data" ;
            :references = "A. Pisano, B. Buongiorno Nardelli, C. Tronconi, R. Santoleri: The
new Mediterranean optimally interpolated pathfinder AVHRR SST Dataset (1982-2012).
```

/Remote Sensing of Environment./ 176 (2016) 107-116, doi:10.1016/j.rse.2016.01.019; http://pathfinder.nodc.noaa.gov and Casey, K.S., T.B. Brandon, P. Cornillon, and R. Evans: The Past, Present and Future of the AVHRR Pathfinder SST Program, in Oceanography from Space: Revisited, eds. V. Barale, J.F.R. Gower, and L. Alberotanza, Springer, 2010. DOI: 10.1007/978-90-481-8681-5_16." ;
:institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem study)" ;
:history = "GOS-CMEMS processor V4: new version" ;

:comment = "WARNING: some applications are unable to properly handle byte values. If Values >127 are encounterd, please subtract 256" ;
:license = "free registration at Copernicus Marine Service (http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
:id = "OISST_HR_REP-GOS-L4-MED-v3.0" ;
:DSD_entry_id = "-GOS-L4HRfnd-MED" ;
:naming_authority = "org.ghrsst" ;
:product_version = "3.0" ;
:uuid = "d18cf538-3619-11f0-a90a-005056be6f27" ;
:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20250521T080136Z" ;
:file_quality_level = 3 ;
:spatial_resolution = "0.05 degree" ;
:start_time = "20250420T190000Z" ;
:time_coverage_start = "20250420T190000Z" ;
:stop_time = "20250421T070000Z" ;
:time_coverage_end = "20250421T070000Z" ;
:software_version = "Copernicus Marine Service HR L4 Processor V.2" ;
:northernmost_latitude = 46.025f ;
:southernmost_latitude = 30.125f ;
:easternmost_longitude = 36.325f ;
:westernmost_longitude = -18.125f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;
:source = "[1982-present] ESA CCI SST v.3.0 L3C product (SST at 0.2m); [1982-2014] Pathfinder-PFV5.3 L3C product (SST skin)" ;
:platform = "NOAA AVHRR, Metop-A and Metop-B AVHRR, (A)ATSR, Sentinel-3A and 3B SLSTR series of sensors" ;
:sensor = "[1982-2018] NOAA AVHRR series of sensors (NOAA-07/08/09/10/11/12/14/15/16/17/18/19), ATSR1, ATSR2, (A)ATSR, MetopA AVHRR; [2017-present] MetopA AVHRR, MetopB AVHRR, SLSTR-3A, SLSTR-3B" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

```
            :standard_name_vocabulary = "NetCDF Climate and Forecast (CF)
Metadata Convention" ;
            :geospatial_lat_units = "degrees_north" ;
            :geospatial_lat_resolution = 0.05f ;
            :geospatial_lon_units = "degrees_east" ;
            :geospatial_lon_resolution = 0.05f ;
            :acknowledgment = "Please acknowledge the use of these data with the
following statement: Generated/provided by Copernicus Marine Service and CNR -
ISMAR ROME. We would also appreciate being informed of any publications." ;
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
}
```

For dataset cmems_obs-sst_med_phy_my_l3s_P1D-m

netcdf \20250421000000-GOS-L3S_GHRSST-SSTfnd-OISST_HR_REP-MED-v02.0-fv04.0

```
{
dimensions:
        time = 1 ;
        lat = 318 ;
        lon = 1089 ;
variables:
        int time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;


                time:calendar = "Gregorian" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:comment = "Includes leap second since 1981" ;
        float lon(lon) ;
                lon:long_name = "longitude" ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
                lon:valid_min = -20.f ;
                lon:valid_max = 40.f ;
        float lat(lat) ;
                lat:long_name = "latitude" ;
```

```
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
            lat:comment = "geographical coordinates, WGS84 projection" ;
            lat:valid_min = 29.f ;
            lat:valid_max = 47.f ;
      short adjusted_sea_surface_temperature(time, lat, lon) ;
            adjusted_sea_surface_temperature:long_name = "adjusted sea surface
foundation_temperature" ;
            adjusted_sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
            adjusted_sea_surface_temperature:type = "foundation" ;
            adjusted_sea_surface_temperature:units = "kelvin" ;
            adjusted_sea_surface_temperature:add_offset = 273.15f ;
            adjusted_sea_surface_temperature:scale_factor = 0.01f ;
            adjusted_sea_surface_temperature:valid_min = -300s ;
            adjusted_sea_surface_temperature:valid_max = 4500s ;
            adjusted_sea_surface_temperature:source = "[1982-present] ESA CCI SST
v.3.0 L3C product (SST at 0.2m); [1982-2014] Pathfinder-PFV5.3 L3C product (SST skin)" ;
            adjusted_sea_surface_temperature:comment = " " ;
            adjusted_sea_surface_temperature:reference = "SLSTRA_D3" ;
            adjusted_sea_surface_temperature:_FillValue = -32768s ;

// global attributes:

            :Conventions = "CF-1.4" ;



            :title = "Mediterranean Sea SST, L3S, Reprocessed using ESA CCI SST v.3.0 and
PFV53 data, 0.05 deg. daily (SST_MED_PHY_L3S_MY_010_042)" ;
            :summary = "[1982-present] Daily gap-free maps (L4) at 0.05deg. x 0.05deg.
horizontal resolution over the Mediterranean Sea. The maps are obtained using
nighttime data extracted from ESA CCI SST v.3.0 and PFV53 data" ;
            :references = "A. Pisano, B. Buongiorno Nardelli, C. Tronconi, R. Santoleri: The
new Mediterranean optimally interpolated pathfinder AVHRR SST Dataset (1982-2012).
/Remote Sensing of Environment./ 176 (2016) 107-116, doi:10.1016/j.rse.2016.01.019;
http://pathfinder.nodc.noaa.gov and Casey, K.S., T.B. Brandon, P. Cornillon, and R.
Evans: The Past, Present and Future of the AVHRR Pathfinder SST Program, in
Oceanography from Space: Revisited, eds. V. Barale, J.F.R. Gower, and L.
Alberotanza, Springer, 2010. DOI: 10.1007/978-90-481-8681-5_16." ;
            :institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem
study)" ;
            :history = "MED-REP-L3S processor, first release" ;
            :comment = "WARNING: the last 5 variable are still in progress - WARNING:
some applications are unable to properly handle byte values. If Values >127 are
encounterd, please subtract 256" ;
            :license = "free registration at Copernicus Marine Service
(http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
            :id = "OISST_HR_REP-GOS-L3S-MED-v4.0" ;
```

:DSD_entry_id = "-GOS-L3S_GHRSST-SSTsubskin" ;
:naming_authority = "org.ghrsst" ;
:product_version = "2.0" ;
:uuid = "d21de7aa-3619-11f0-a1e4-005056be6f27" ;
:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20250421T080137Z" ;
:file_quality_level = 3 ;
:spatial_resolution = "0.05 degree" ;
:start_time = "20250420T190000Z" ;
:time_coverage_start = "20250420T190000Z" ;
:stop_time = "20250421T070000Z" ;
:time_coverage_end = "20250421T070000Z" ;
:software_version = "Copernicus Marine Service HR L3 Processor V.1.0" ;
:northernmost_latitude = 46.025f ;
:southernmost_latitude = 30.125f ;
:easternmost_longitude = 36.325f ;
:westernmost_longitude = -18.125f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;

:source_data = "code: ESA CCI SST v.3.0 :1, Pathfinder-PFV5.3 L3C :2" ;
:source = "code: ESA CCI SST v.3.0 :1, Pathfinder-PFV5.3 L3C :2" ;
:sensor = "[1982-2018] NOAA AVHRR series of sensors (NOAA-07/08/09/10/11/12/14/15/16/17/18/19), ATSR1, ATSR2, (A)ATSR, MetopA AVHRR; [2017-present] MetopA AVHRR, MetopB AVHRR, SLSTR-3A, SLSTR-3B" ;
:platform = "NOAA AVHRR, Metop-A AVHRR, Metop-B AVHRR, (A)ATSR, Sentinel-3A and 3B SLSTR series of sensors" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lat_resolution = 0.05f ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lon_resolution = 0.05f ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We would also appreciate being informed of any publications." ;
:creator_name = "ISMAR - Institute of Marine Sciences (CNR - ISMAR - GOS - Rome)" ;
:creator_email = "gsdk@artov.ismar.cnr.it" ;
:creator_url = "http://gosweb.artov.ismar.cnr.it/" ;
:project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
:publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
:publisher_url = "http://marine.copernicus.eu/" ;

```
        :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
        :processing_level = "L3S" ;
        :cdm_data_type = "grid" ;
        :distribution_statement = "See Copernicus Marine Service Data License" ;
}
```